ACT MATH • UNIT 0.01

ACT Math Diagnostic & Pacing Strategy Drill

PRACTICE WORKSHEET

60-Minute Pacing & Core Readiness • Official ACT Math Curriculum

Name: _____ **Date:** _____ **Score:** _____ / 25

ACT Math Diagnostic & Pacing Strategy Drill Practice Exercises

1 MCQ

For the imaginary unit i, where $i^2 = -1$, which of the following complex numbers is equal to $(5 - 3i) - (2 - 8i)$?

A $3 + 5i$

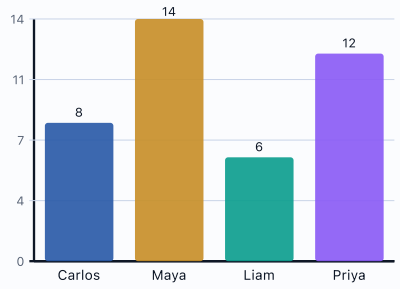
B $3 - 11i$

C $7 - 11i$

D $3 - 5i$

2 MCQ

The bar graph below shows the number of hours four students volunteered at a community center last week. What percent of the total volunteer hours logged by the four students was logged by Maya?



A 14%

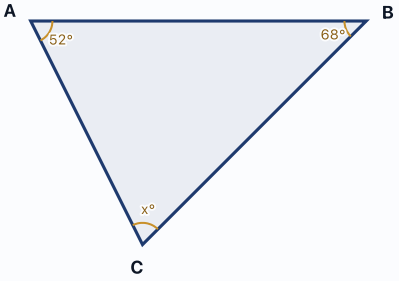
B 28%

C 35%

D 42%

3 MCQ

In $\triangle ABC$ shown below, the measure of $\angle A$ is 52° and the measure of $\angle B$ is 68°. What is the value of x?



A 58

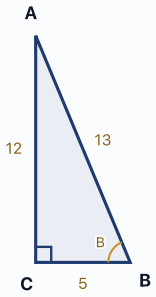
B 60

C 70

D 120

4 MCQ

In the right triangle $\triangle ABC$ shown below, the right angle is at vertex C. The lengths of the sides are given in inches: $AC = 12$, $BC = 5$, and $AB = 13$. What is the value of $\cos B$?



A $\frac{5}{13}$

B $\frac{12}{13}$

C $\frac{5}{12}$

D $\frac{12}{5}$

5 MCQ

In the standard (x, y) coordinate plane, vector $\mathbf{u} = \langle 2, -5 \rangle$ and vector $\mathbf{v} = \langle k, 3 \rangle$, where $k > 0$. If the magnitude of the vector sum $\mathbf{u} + \mathbf{v}$ is $\sqrt{29}$, what is the value of k?

A 3

B 7

C $\sqrt{21}$

D 5

6 MCQ

Which of the following expressions is equivalent to $(5x^2 - 2x + 7) - (3x^2 - 6x - 4)$?

A $2x^2 - 8x + 3$

B $2x^2 - 8x + 11$

C $2x^2 + 4x + 3$

D $2x^2 + 4x + 11$

7 MCQ

In the standard (x, y) coordinate plane, what are the coordinates of the midpoint of a line segment with endpoints $(-3, 8)$ and $(5, -2)$?

A $(1, 3)$

B $(4, -5)$

C $(2, 6)$

D $(3, 1)$

8 MCQ

If $f(x) = 3x^2 - 5x + 4$, what is the value of $f(-2)$?

A -18

B 2

C 6

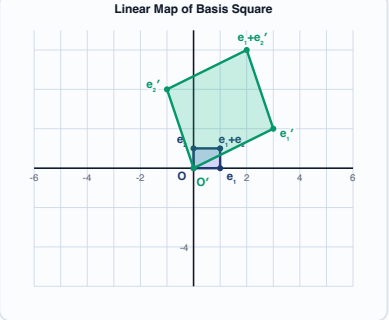
D 26

9 MCQ

In the standard (x, y) coordinate plane, the complex number $z = x + yi$ satisfies the equation

$$z(2 - i) = 15 + 5i$$

where $i^2 = -1$, and x and y are real numbers. Vector $\mathbf{v}$ is defined by $\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$. If matrix $A = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}$, what is the magnitude of the vector $A\mathbf{v}$?



A $10\sqrt{5}$

B $10\sqrt{10}$

C 40

D $5\sqrt{2}$

10 MCQ

In the standard (x, y) coordinate plane, vector $\mathbf{u} = \langle 2, -5 \rangle$ and vector $\mathbf{v} = \langle -3, 4 \rangle$. What is the vector $2\mathbf{u} + \mathbf{v}$?

A $\langle -1, -1 \rangle$

B $\langle 1, -1 \rangle$

C $\langle 1, -6 \rangle$

D $\langle 7, -14 \rangle$

11 MCQ

A bag contains 8 red marbles, 6 green marbles, and 10 blue marbles. If 1 marble is chosen at random from the bag, what is the probability that the marble is NOT red?

A $\frac{1}{3}$

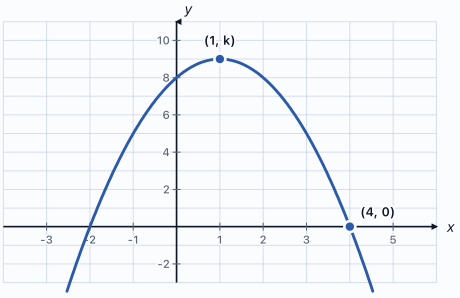
B $\frac{1}{2}$

C $\frac{3}{4}$

D $\frac{2}{3}$

12 MCQ

The graph of the quadratic function $f(x) = -x^2 + bx + c$ is shown in the standard (x, y) coordinate plane below. The vertex of the parabola is at $(1, k)$, and the graph passes through the point $(4, 0)$. What is the value of k?



A 9

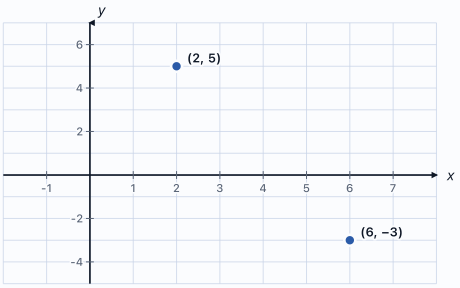
B 6

C 8

D 12

13 MCQ

In the standard (x, y) coordinate plane, line l contains the points $(2, 5)$ and $(6, -3)$. Line k is perpendicular to the segment connecting these two points and passes through its midpoint. What is the y-intercept of line k?



A 9

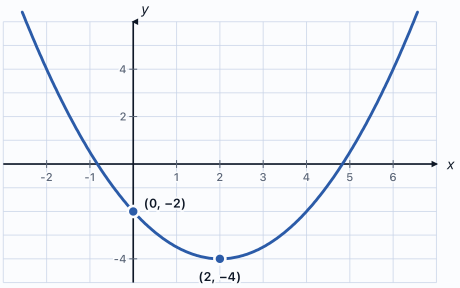
B -1

C -6

D 3

14 MCQ

The graph of the quadratic function $f(x) = a(x-h)^2 + k$ is shown in the standard (x, y) coordinate plane below, with vertex $(2, -4)$ and y-intercept $(0, -2)$. If the point $(6, c)$ lies on the graph, what is the value of c?



A 4

B 2

C 8

D 12

15 MCQ

On the number line shown below, point A is located at -16 and point B is located at 24. Point P lies on the line segment connecting A and B such that the ratio of the length of $\overline{AP}$ to the length of $\overline{PB}$ is $3 : 2$. What is the coordinate of point P?



A 0

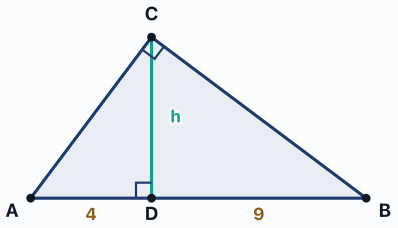
B 8

C 4

D 16

16 MCQ

In the figure below, $\triangle ABC$ is a right triangle with right angle at C, and altitude $\overline{CD}$ of length h is perpendicular to hypotenuse $\overline{AB}$. If $AD = 4$ and $DB = 9$, what is the area of $\triangle ABC$?



A 27

B 36

C 39

D 78

17 MCQ

A list consists of 7 distinct integers. The arithmetic mean of the 7 integers is 24, and the median is 22. If the 3 smallest integers are 15, 18, and 20, what is the maximum possible value of the largest integer in the list?

A 47

B 48

C 49

D 46

18 MCQ

The function $f(x) = a\sin(bx) + c$ has a maximum value of 9, a minimum value of -1, and a period of $\frac{\pi}{2}$. If $a > 0$ and $b > 0$, what is the value of $a + b + c$?

A 9

B 13

C 11

D 18

19 MCQ

The quadratic function $f(x) = x^2 - 2kx + (3k + 10)$ has 2 distinct real zeros. If both zeros are strictly greater than 2, how many integer values of k are possible?

A 7

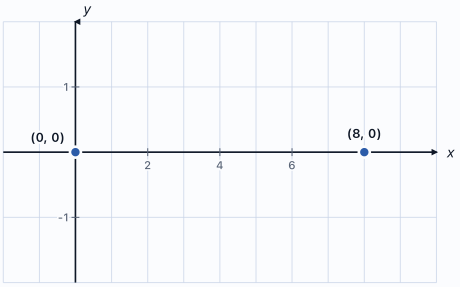
B 9

C 11

D 8

20 MCQ

In the standard (x, y) coordinate plane, a triangle has vertices at $O(0, 0)$, $A(8, 0)$, and $B(2, 6)$. A horizontal line with equation $y = c$, where $0 < c < 6$, divides the interior of $\triangle OAB$ into two regions of equal area. What is the value of c?



A $6 - 2\sqrt{3}$

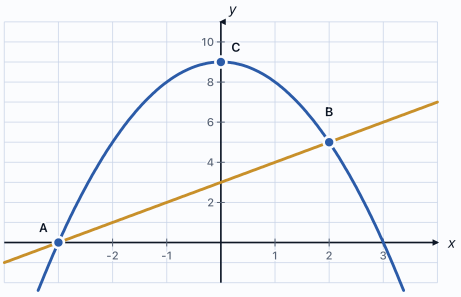
B 3

C $3\sqrt{2}$

D $6 - 3\sqrt{2}$

21 MCQ

In the standard (x, y) coordinate plane below, the graph of the quadratic function $f(x) = 9 - x^2$ and the line $g(x) = x + 3$ intersect at points A and B, where A has a negative x-coordinate and B has a positive x-coordinate. Point C is the vertex of the parabola $y = f(x)$. What is the area, in square coordinate units, of $\triangle ABC$?



A 12.5

B 15

C 22.5

D 30

22 MCQ

Every student in the graduating class of Oakridge High School plans to do exactly one of the following after graduation: attend a 2-year college, attend a 4-year university, or enter the workforce directly.

• The ratio of the number of students planning to attend a 2-year college to the number of students planning to attend a 4-year university is $3:5$.

• Of the students planning to attend a 2-year college, exactly 80% will attend an in-state college.

• Of the students planning to attend a 4-year university, exactly 72% will attend an in-state university.

• The total number of students planning to attend an in-state college or university accounts for exactly 45% of the entire graduating class.

If the number of students in every category and subcategory must be a positive integer, what is the minimum possible number of students in the graduating class?

A 200

B 100

C 120

D 40

23 MCQ

An isosceles trapezoid has parallel bases of lengths 4 cm and 16 cm. A circle is inscribed inside the trapezoid such that it is tangent to all four sides. What is the area, in square centimeters, of the region inside the trapezoid that lies outside the circle?

A $64-16\pi$

B $80-16\pi$

C $80-25\pi$

D $100-25\pi$

24 MCQ

A box contains 10 cards, each labeled with a distinct integer from 1 through 10. If 3 cards are drawn at random from the box without replacement, what is the probability that the sum of the numbers on the 3 drawn cards is divisible by 3?

A $\frac{7}{20}$

B $\frac{1}{4}$

C $\frac{3}{10}$

D $\frac{1}{3}$

25 MCQ

In $\triangle ABC$, side AB has length 6, side AC has length 10, and the measure of $\angle BAC$ is 120°. The angle bisector of $\angle BAC$ intersects side BC at point D. What is the length of segment AD?

A $\frac{30}{7}$

B $\frac{15}{4}$

C $\frac{15\sqrt{3}}{4}$

D $\frac{15}{2}$